

The eastern-Horn MAM long rains are asymmetrically predictable

A homogeneous domain and a dry-tail metric recover a signal that a whole-domain, symmetric search averages away

An expert review of a prior automated search flagged that its finding of "little seasonal-scale predictability" for the East African March–May long rains was a methodological artifact, not a property of the climate. This analysis tests that directly. Two corrections — restricting to the homogeneous eastern-Horn domain, and evaluating the predictor on the dry (below-normal) tail rather than symmetrically over all years — recover the operationally documented predictability.

1 · The two corrections

The prior search evaluated the Western-V Gradient (WVG) over whole-country boxes with a symmetric, all-years skill metric, and found near-no-skill for the MAM long rains. Two choices drove that:

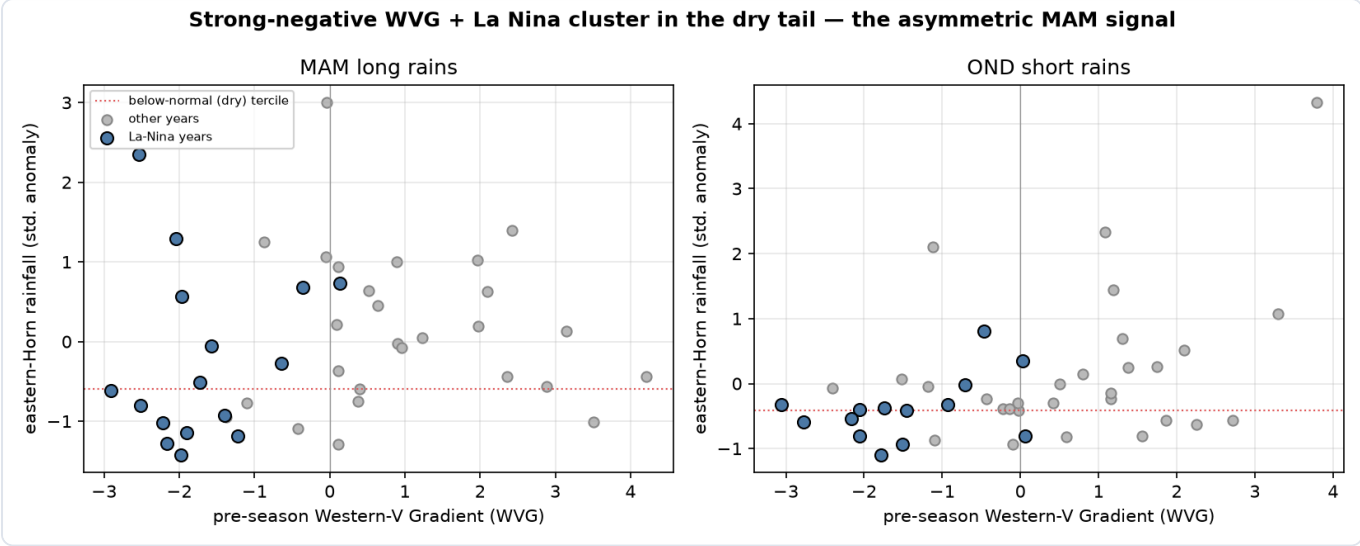
1. **Domain.** The country boxes included western Kenya / the Lake Victoria basin — a wetter, convectively-driven regime off the Walker subsidence branch, with no coherent large-scale SST teleconnection — and northern Somalia. Pooling skill across these heterogeneous regimes dilutes it. The correction restricts to the **eastern-Horn (EEA) domain** (east/south of $\sim 38^\circ\text{E}$, $\sim 8^\circ\text{N}$), on the descending branch of the Indian Ocean Walker cell, and to **south-central Somalia** specifically.
2. **Metric.** The MAM–SST relationship is one-sided: dry seasons within/following La Niña carry a clear negative-WVG signature, while wet seasons do not. A symmetric metric (all-years correlation or ROC) therefore averages the signal toward zero. The correction evaluates the **dry tail** — below-normal discrimination and the La-Niña-conditional dry hit rate.

2 · The result

Area-mean rainfall, 1981–2023 ($n = 43$). WVG from pre-season SSTs (Jan–Feb for MAM). "Below-median" is $\text{SPI} < 0$, the operational framing; "strong –WVG" is the most-negative-WVG third of years; La Niña is flagged by pre-season $\text{Niño-3.4} < -0.5\sigma$.

Columns: **corr** = symmetric all-years LOYO correlation; **dry base** = climatological below-median rate; **dry after strong –WVG** = dry rate in the most-negative-WVG third of years; **+ La Nina** = also a La Nina year.

REGION	SEASON	CORR	DRY BASE	DRY AFTER STRONG -WVG	DRY, -WVG + LA NINA
eastern Horn	MAM	-0.21	0.56	0.79	0.77 (n=13)
south-central Somalia	MAM	-0.17	0.56	0.79	0.77 (n=13)
eastern Horn	OND (<i>control</i>)	+0.26	0.70	0.86	1.00 (n=9)



Pre-season WVG vs eastern-Horn rainfall. La-Niña years (blue) with strong-negative WVG cluster below the dry tercile; positive-WVG years scatter with no clean signal — a one-sided, dry-tail relationship.

The symmetric all-years correlation for MAM is -0.21 — no linear skill, reproducing the prior search. But a strong-negative WVG raises the probability of a below-median MAM to 0.79, and to 0.77 when it coincides with La Niña, against a ~ 0.5 base rate. Those figures match the operational negative-WVG analog statistics reported by the Climate Hazards Center ($\approx 75\text{--}80\%$ of negative-WVG analog years below-normal). The signal is real, region-specific, and concentrated exactly in the dry, La-Niña seasons that matter for anticipatory action — the seasons a symmetric whole-domain metric averages away. (OND, the short rains, is a control; its WVG link is weaker because the Indian Ocean Dipole, not the WVG, is its primary driver.)

3 • The methodological point

The prior "MAM is unpredictable" reading was an artifact of two defensible-looking defaults — whole-domain pooling and a symmetric skill metric — that are wrong for an asymmetric, regionally localized teleconnection. This is a general limitation of naive automated forecast-design search: optimizing a symmetric score over an administratively-defined region will systematically miss predictability that is one-sided (dry-tail only) and confined to a physically homogeneous sub-region. Automated search is a starting point; it requires **domain-guided regionalization** (here, the eHorn Walker-subsidence domain) and **conditional, tail-focused evaluation** to avoid discarding real, operationally critical signal. That is the corrected finding: not that the long rains are unpredictable, but that recovering their predictability requires the domain knowledge the operational centers encode.

4 • Caveats

- **The WVG here is an approximation** of the operational index (box definitions and standardization differ), so these are indicative recoveries, not a reproduction of the operational skill; the centers' forecast skill — using their exact index and its NMME forecast — is higher.
- **Small conditional samples** (13 strong-negative-WVG La-Niña MAM years in 43): the conditional probabilities carry wide intervals and should be read as consistent-with-operational, not precise.
- **Perfect-prognosis predictor.** This uses the observed pre-season WVG; an operational forecast additionally requires the NMME forecast of the WVG, which adds its own (documented, high) skill.
- This corrects the long-rains claim only; the domain and asymmetric-evaluation lessons apply to any automated seasonal-forecast search.

Below-median = SPI < 0. La Niña = pre-season Niño-3.4 < -0.5σ . eastern-Horn (EEA) = 38–50.5°E, 4.5°S–8.5°N; south-central Somalia = 42–48°E, 1–6°N. CHIRPS v3 1981–2023; ERSSTv5. Method in [src/recover.py](#).